Experiment 16. EDA: quantitative and qualitative analysis

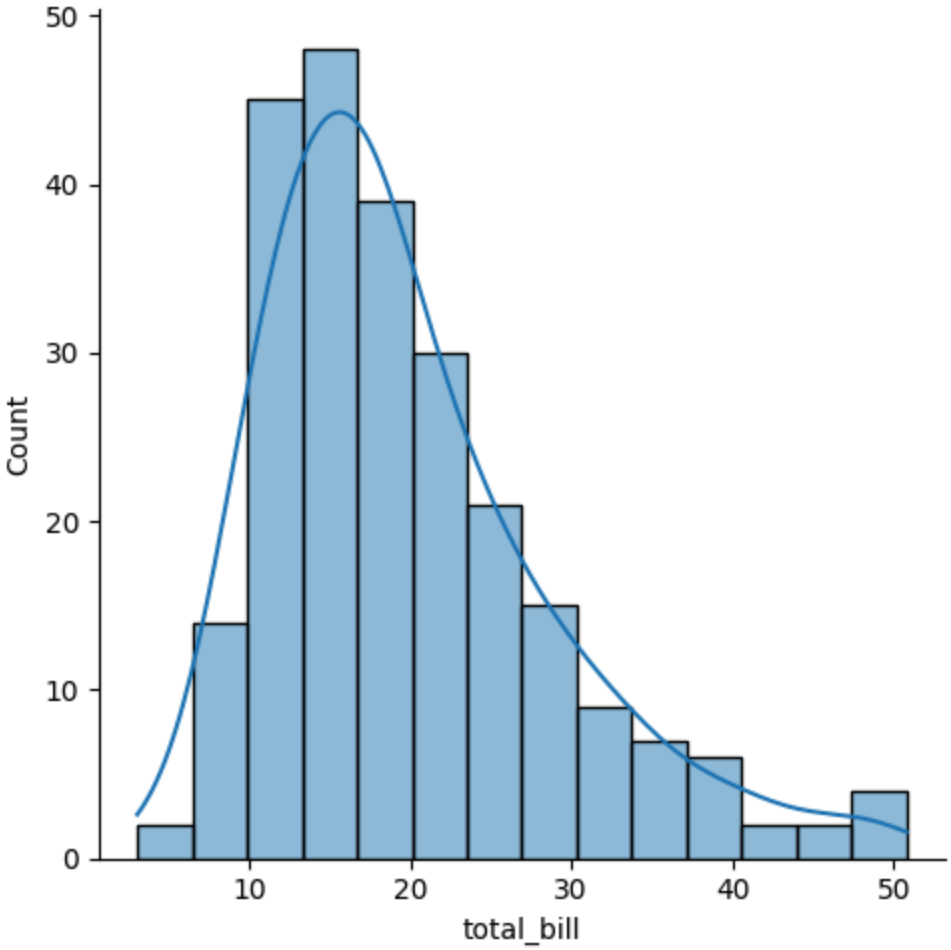
In [1]:
```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
tips=sns.load_dataset('tips')
print(tips.head())
sns.displot(tips.total_bill,kde=True)
sns.displot(tips.total_bill,kde=False)
sns.jointplot(x=tips.tip,y=tips.total_bill)
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
sns.pairplot(tips)
tips.time.value_counts()
sns.pairplot(tips,hue='time')
sns.pairplot(tips,hue='day')
sns.heatmap(tips.corr(numeric_only=True),annot=True)
sns.boxplot(tips.total_bill)
sns.boxplot(tips.tip)
sns.countplot(tips.day)
sns.countplot(tips.sex)
tips.sex.value_counts().plot(kind='pie')
tips.sex.value_counts().plot(kind='bar')
sns.countplot(tips[tips.time=='Dinner']['day'])
```

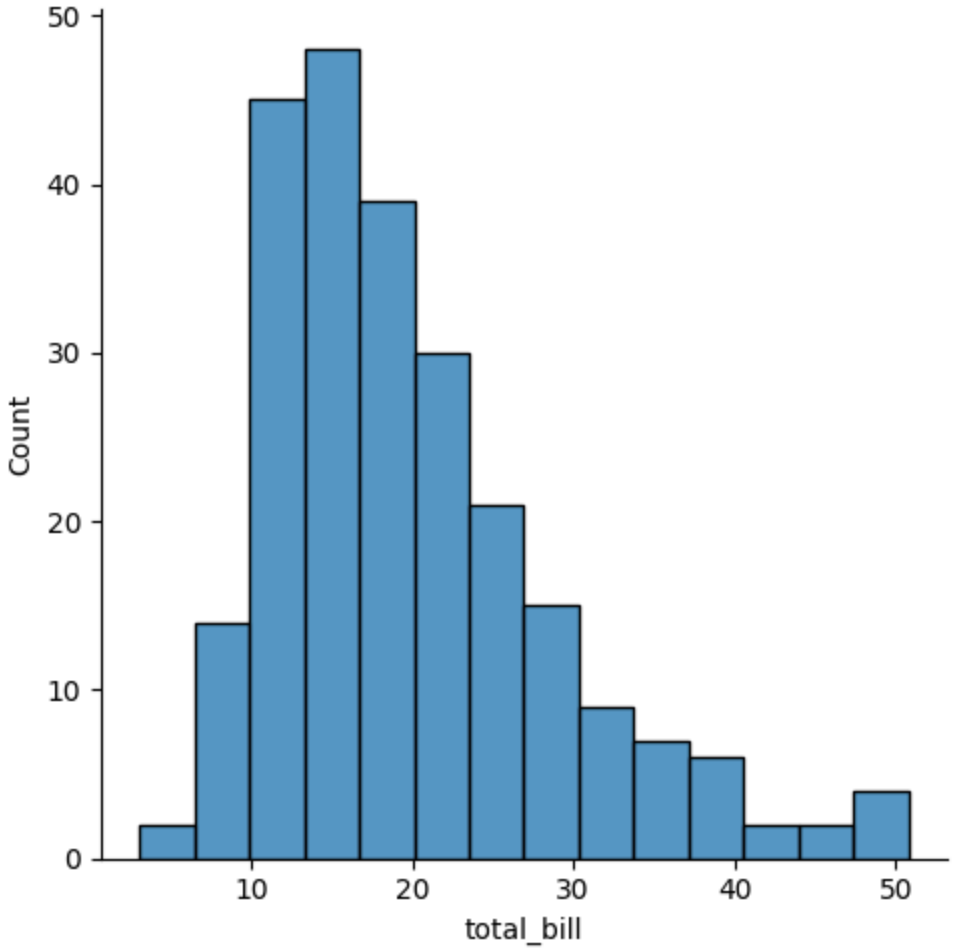
```
   total_bill   tip     sex smoker  day    time  size
0       16.99  1.01  Female     No  Sun  Dinner     2
1       10.34  1.66    Male     No  Sun  Dinner     3
2       21.01  3.50    Male     No  Sun  Dinner     3
3       23.68  3.31    Male     No  Sun  Dinner     2
4       24.59  3.61  Female     No  Sun  Dinner     4
```

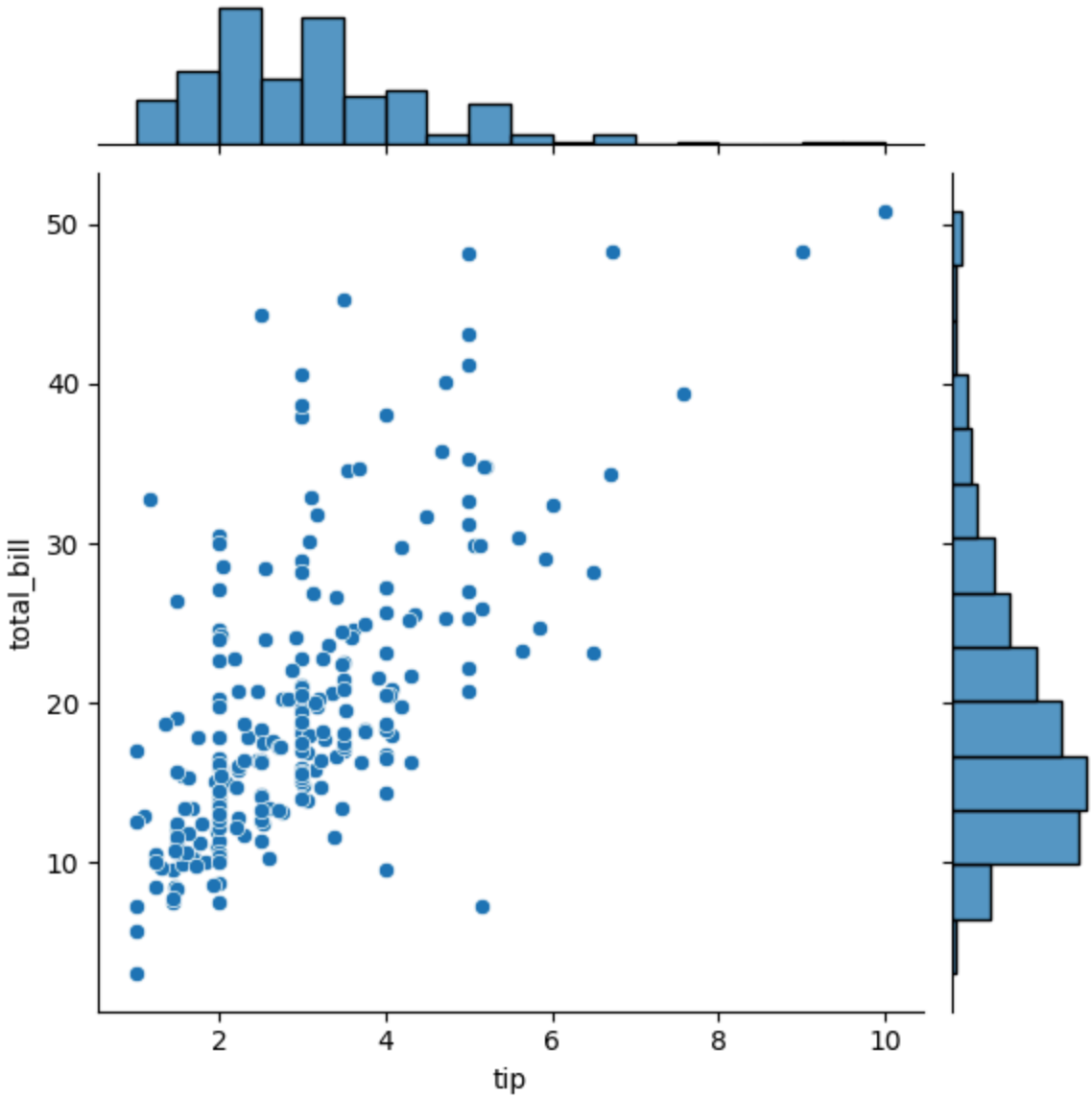
```
WARNING:matplotlib.axes._base:Ignoring fixed y limits to fulfill fixed data aspect with adju
stable data limits.
/usr/local/lib/python3.13/dist-packages/seaborn/categorical.py:383: UserWarning: Attempting
to set identical low and high ylims makes transformation singular; automatically expanding.
  ax.set_ylim(n - .5, -.5, auto=None)
```

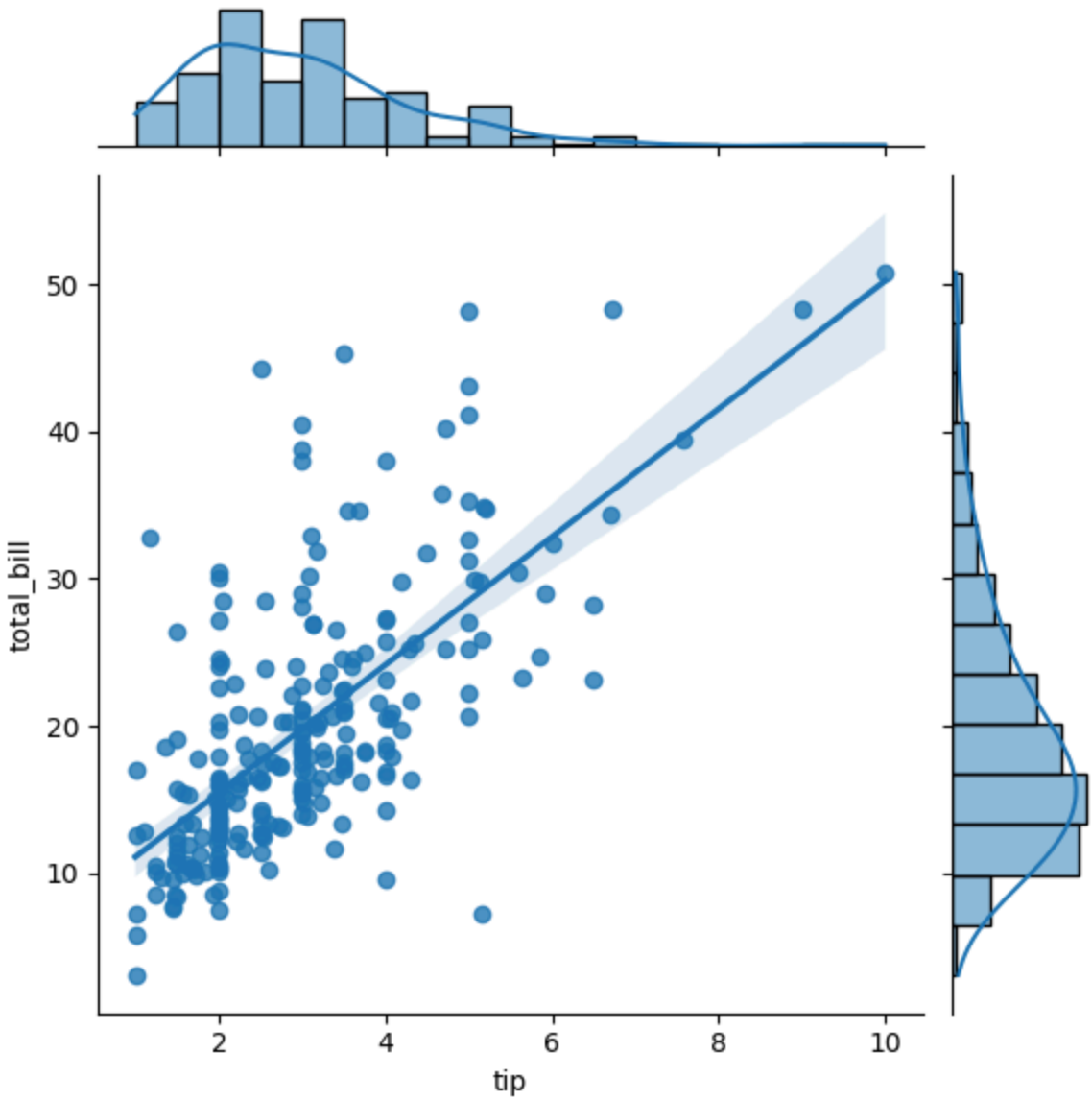
Out[1]:
```
<Axes: xlabel='sex', ylabel='count'>
```

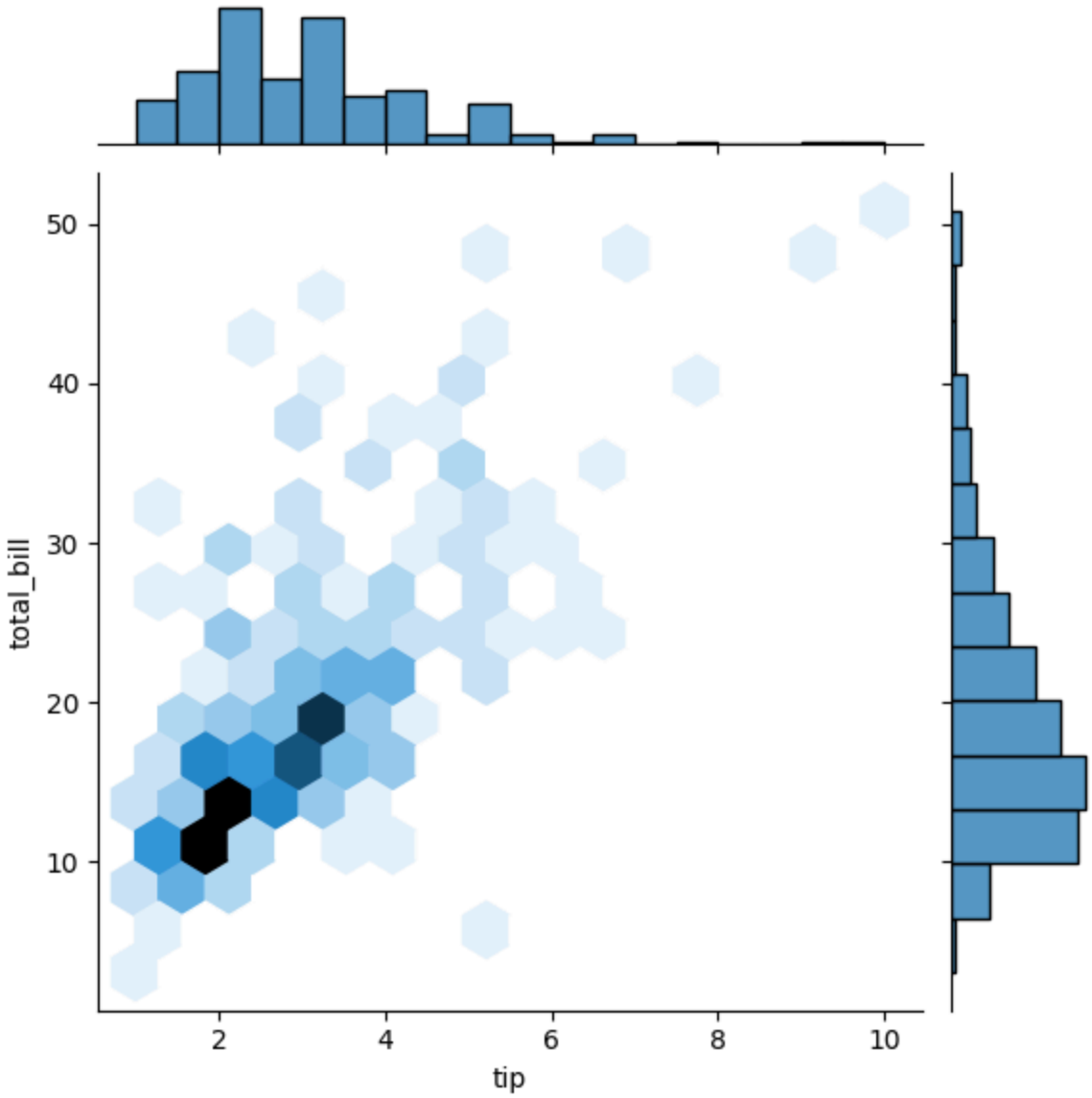
```
WARNING:matplotlib.axes._base:Ignoring fixed y limits to fulfill fixed data aspect with adju
stable data limits.
```

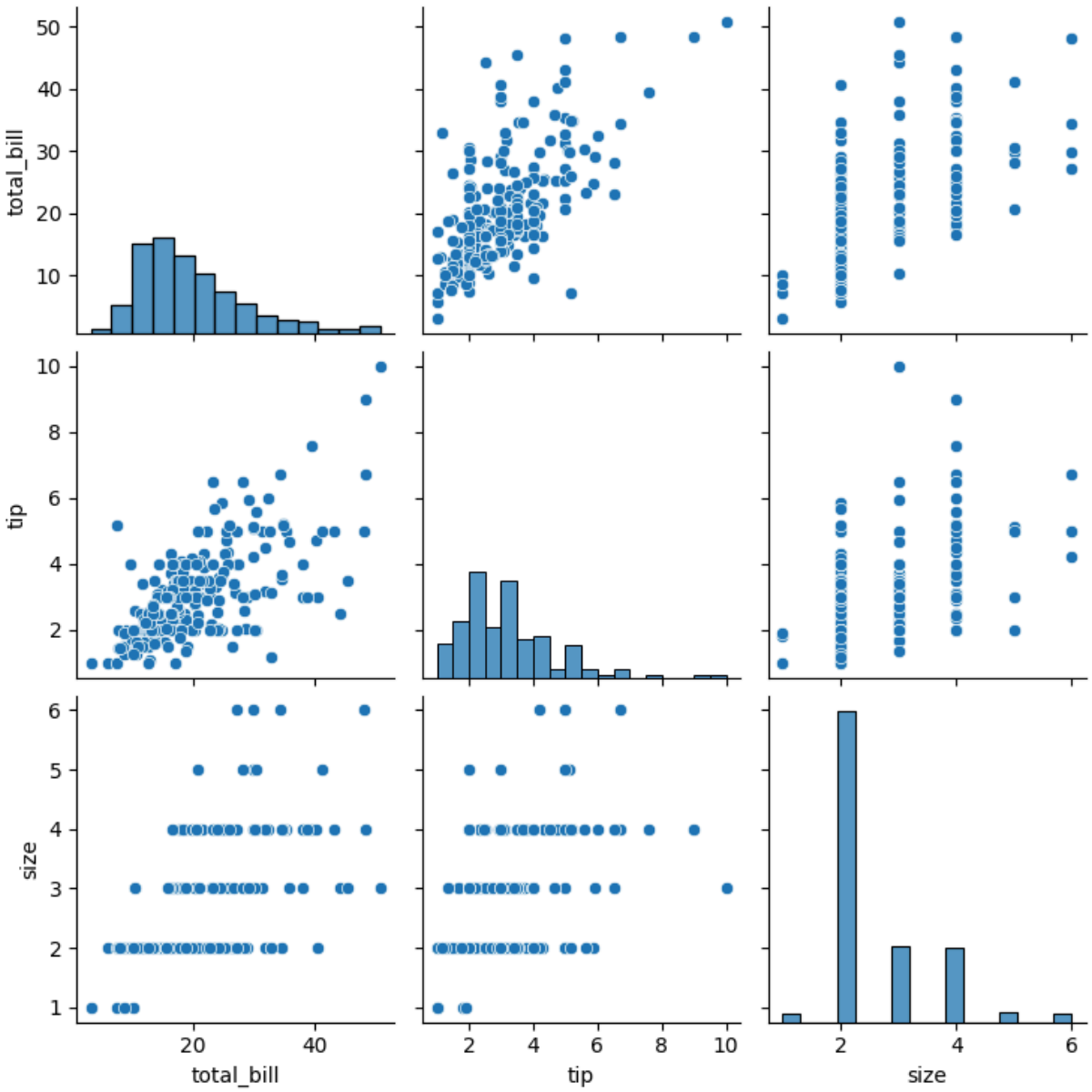

50
40
30
20
10
0
Count
10
20
30
40
50
total_bill

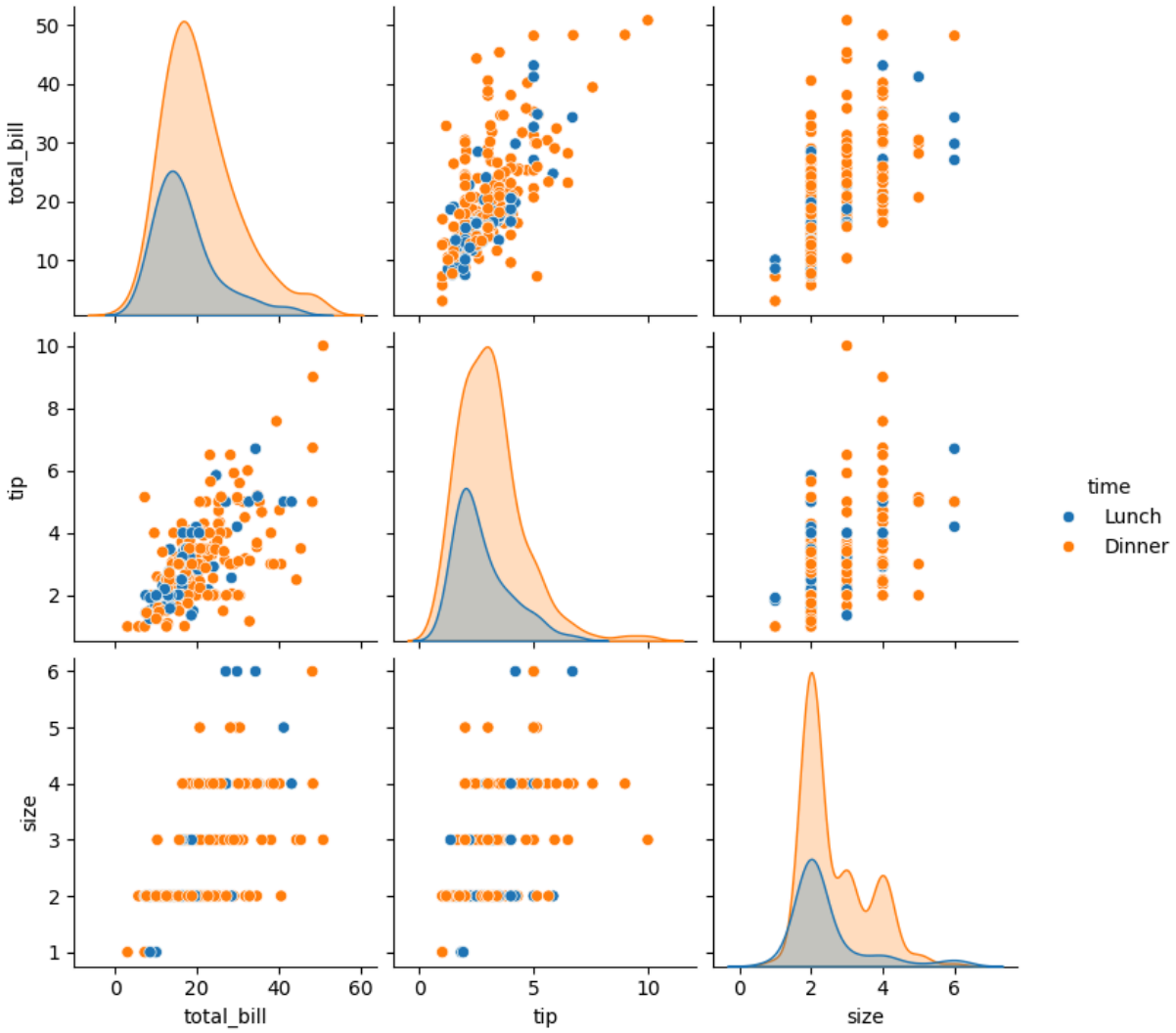

50
40
30
20
10
0
Count
10
20
30
40
50
total_bill

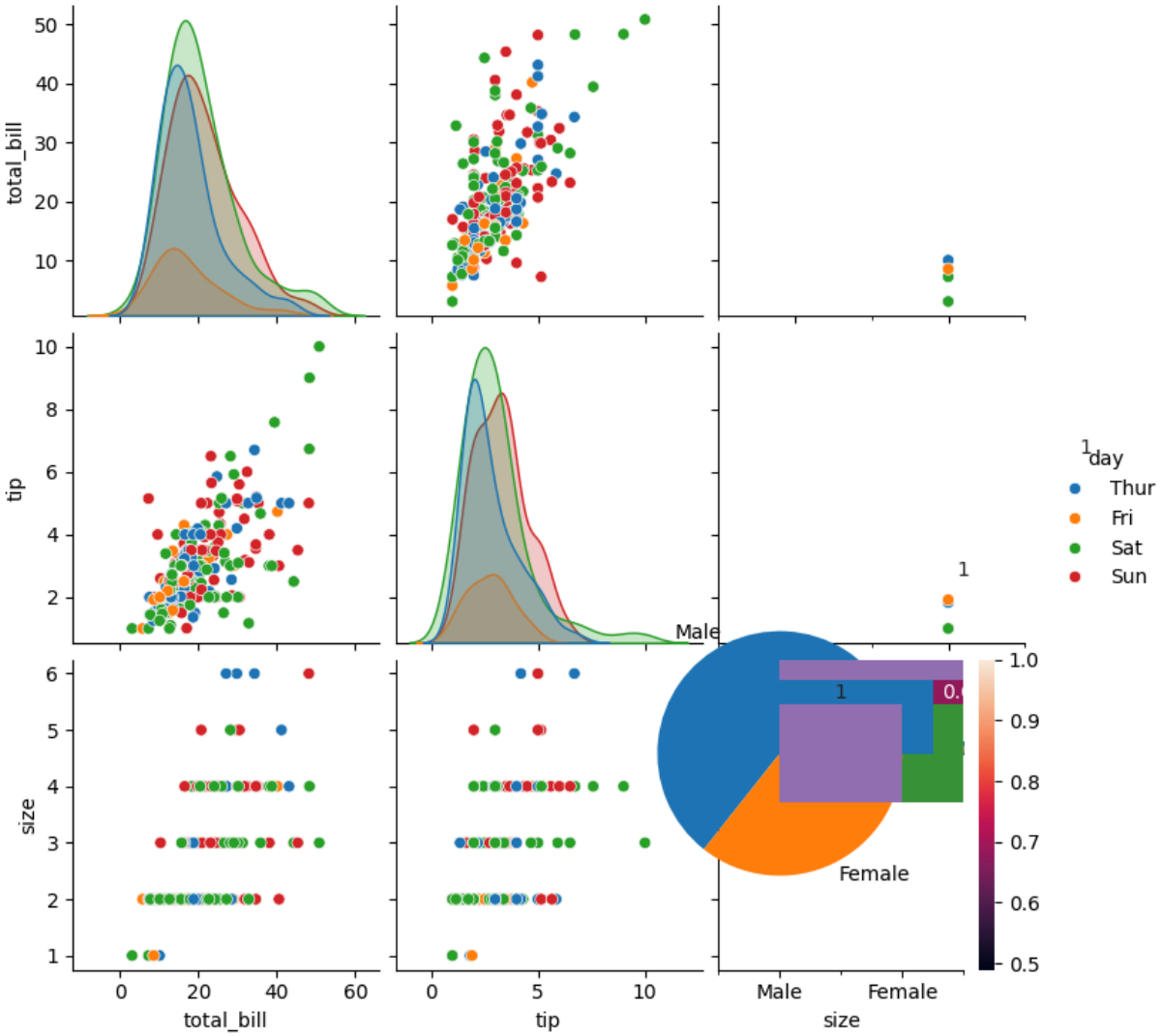

total_bill
tip
10
20
30
40
50
2
4
6
8
10


total_bill
tip
10
20
30
40
50
2
4
6
8
10


total_bill
tip
50
40
30
20
10
2
4
6
8
10


total_bill
tip
size
50
40
30
20
10
10
8
6
4
2
6
5
4
3
2
1
20
40
2
4
6
8
10
2
4
6


total_bill
tip
size
time
Lunch
Dinner


total_bill
tip
size
day
Thur
Fri
Sat
Sun
Male
Female
1
0.5
0.6
0.7
0.8
0.9
1.0